

Artificial Intelligence

AI2002

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Hill Climbing Algorithm

- Q1
- We are given a grid where each cell has a value (height).
The goal is to reach the **highest value (peak)** using **steepest ascent hill climbing**.

	C1	C2	C3	C4
R1	1	2	3	4
R2	2	5	6	5
R3	1	3	9	4
R4	0	2	3	1

- From any cell, you can move to: Up, Down, Left, Right (no diagonals)
- At each step: Choose the **neighbor with the highest value** (steepest ascent)
- Tie breaking rule: Pick the higher Row
- Initial State (R2, C1) = 2
- Does the algorithm reach global maxima in the given example?
- Find the path

	C1	C2	C3	C4
R1	1	2	3	4
R2	2	5	6	5
R3	1	3	9	4
R4	0	2	3	1

- Start (R2,C1) = 2
- Neighbours – {(R1, C1) = 1 (R3, C1) = 1 **(R2, C2) = 5**}
- From (R2, C2) = 5
- Neighbours – {(R1, C2) = 2 (R3, C2) = 3 (R2, C1) = 2 **(R2, C3) = 6**}
- From (R2, C3) = 6
- Neighbours – {(R1, C3) = 3 **(R3, C3) = 9** (R2, C2) = 5 (R2, C4) = 5}
- From (R3, C3) = 9
- Neighbours – {(R2, C3) = 6 (R4, C3) = 3 (R3, C2) = 3 (R3, C4) = 4}
- No neighbor has a higher value than 9
- **Path followed:**
(R2, C1) → (R2, C2) → (R2, C3) → (R3, C3) **Peak reached: 9 at (R3, C3) -> Global Maxima**

Genetic algorithms

- Q2
- A population of four chromosomes is given below. Each chromosome consists of 8 bits.

Chromosome	Fitness
C1: 11001010	5
C2: 10110100	8
C3: 01101101	3
C4: 11100011	10

- Given is the list of random numbers that can be used throughout the algorithm.
0.72, 0.15, 0.63, 0.48, 0.91, 0.27, 0.36, 0.84, 0.05, 0.58, 0.77, 0.19
- (a) Perform **fitness proportionate selection** to select **two chromosomes** using the given random numbers and show which chromosomes are selected
- (b) Assume the two selected chromosomes undergo **two-point crossover**. Determine crossover points using the random numbers and show the **resulting offspring**
- (c) Apply **two-point mutation** on the chromosome: **10011001**. Show the resulting chromosome

(a)

- Total chromosomes = 4 Length N=8
- Random numbers = {0.72, 0.15, 0.63, 0.48, 0.91, 0.27, 0.36, 0.84, 0.05, 0.58, 0.77, 0.19}

- Total fitness = 5 + 8 + 3 + 10 = 26

Chromosome	Fitness	Probability (p)	Ranges
C1: 11001010	5	$5/26 = 0.19$	0-0.19
C2: 10110100	8	$8/26 = 0.31$	0.19-0.50
C3: 01101101	3	$3/26 = 0.12$	0.50-0.62
C4: 11100011	10	$10/26 = 0.38$	0.62-1.00

- R1 = 0.72
 - Falls in range 0.62 – 1.00 → C4 will be selected
- R2 = 0.15
 - Falls in range 0 – 0.19 → C1 will be selected

(b)

- Random numbers = {0.72, 0.15, 0.63, 0.48, 0.91, 0.27, 0.36, 0.84, 0.05, 0.58, 0.77, 0.19}
- Selected Chromosomes
 - C4: 11100011
 - C1: 11001010
- To determine positions of two point crossover we use
 - **Position** = $\lceil R \times N \rceil$, where $N=8$
 - $R = 0.63 \rightarrow \lceil 0.63 \times 8 \rceil = \lceil 5.04 \rceil = 6$
 - $R = 0.48 \rightarrow \lceil 0.48 \times 8 \rceil = \lceil 3.84 \rceil = 4$
- The bits between 4-6 will be exchanged
 - Index: 1 2 3 4 5 6 7 8
 - C4: 1 1 1 **0 0 0** 1 1
 - C1: 1 1 0 **0 1 0** 1 0
- Offspring
 - C5: 11101011
 - C6: 11000010

(c)

- Random numbers = {0.72, 0.15, 0.63, 0.48, 0.91, 0.27, 0.36, 0.84, 0.05, 0.58, 0.77, 0.19}
- Given chromosome: 10011001
- Finding mutation positions using **Position = $\lceil R \times N \rceil$** , where $N=8$
 - $R = 0.91 \rightarrow \lceil 0.91 \times 8 \rceil = \lceil 7.28 \rceil = 8$
 - $R = 0.27 \rightarrow \lceil 0.27 \times 8 \rceil = \lceil 2.16 \rceil = 3$
- Bits 3 and 8 will be flipped in the given chromosome
 - Index: 1 2 3 4 5 6 7 8
 - Original: 1 0 **0** 1 1 0 0 **1**
 - Mutated: 1 0 **1** 1 1 0 0 **0**

Linear Regression

- Q3

- A fitness trainer wants to analyze how daily exercise affects body weight. She records the following data for 5 individuals:

Person	Exercise Time (minutes/day) (X)	Weight (kg) (Y)
1	10	85
2	20	80
3	30	75
4	40	70
5	50	65

- (a) Calculate Slope b_1 , Intercept b_0 and Write the regression equation
- (b) Compute: SSE, SST and SSR
- (c) Evaluate: Coefficient of determination r^2 and Correlation coefficient r_{xy}
- (d) Predict the weight of a person who exercises **35 minutes/day**
- (e) Interpret the meaning of the slope if its value is -0.5
- (f) What will be the change in the predicted value $\hat{y}$ if x increases by 5 units?
- (g) For an input $x = 15$, the actual value is $y = 83$. Compute the prediction error

(a)

- Computing means
- Calculate slope

$$\bar{x} = \frac{10 + 20 + 30 + 40 + 50}{5} = 30$$

$$\bar{y} = \frac{85 + 80 + 75 + 70 + 65}{5} = 75$$

$$b_1 = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sum(x - \bar{x})^2}$$

$$b_1 = \frac{-500}{1000} = -0.5$$

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$	$(x - \bar{x})^2$
10	85	-20	10	-200	400
20	80	-10	5	-50	100
30	75	0	0	0	0
40	70	10	-5	-50	100
50	65	20	-10	-200	400

- Calculate intercept
- Regression equation

$$b_0 = \bar{y} - b_1\bar{x} = 75 - (-0.5 \times 30) = 75 + 15 = 90$$

$$\hat{y} = 90 - 0.5x$$

(b)

x	y	$y - \bar{y}$	$(y - \bar{y})^2$	$\hat{y} = 90 - 0.5x$	$(\hat{y} - \bar{y})^2$	$(y - \hat{y})^2$
10	85	10	100	85	100	0
20	80	5	25	80	25	0
30	75	0	0	75	0	0
40	70	-5	25	70	25	0
50	65	-10	100	65	100	0
SST = 250				SSR = 250		SSE = 0

(c)

$$r^2 = \frac{SSR}{SST} = \frac{250}{250} = 1$$

$$r = -1 \quad (\text{negative because slope is negative})$$

(d) At x=35 minutes, predicted weight is 72.5 kgs

$$\hat{y} = 90 - 0.5(35) = 90 - 17.5 = 72.5$$

(e) $\hat{y} = 90 - 0.5x$

For every 1-minute increase in exercise, weight decreases by 0.5 kg suggesting a strong negative linear relationship as $r=-1$

(f) Change in $\hat{y}$ if x increases by 5 units

$$\Delta\hat{y} = -0.5 \times 5 = -2.5$$

- $\hat{y}$ decreases by 2.5 units

(g) Prediction error for $x = 15$, $y = 83$

$$\hat{y} = 90 - 0.5(15) = 90 - 7.5 = 82.5$$

$$\text{Error} = y - \hat{y} = 83 - 82.5 = 0.5$$

Perceptron Learning Rule

- Q4

- Activation function:

$$f(\text{net}) = \begin{cases} 1 & \text{if net} \geq 0 \\ -1 & \text{if net} < 0 \end{cases}$$

- Bias input: $x_0=1$, weight = w_0

- Initial weights: $w_0=0$, $w_1=0$, $w_2=0$

- Learning rate: $\eta=1$

- $\text{net} = w_0x_0 + w_1x_1 + w_2x_2$

- Update rule:

- $w_i^{\text{new}} = w_i^{\text{old}} + \eta \cdot y \cdot x_i$

x1	x2	y (label)
1	1	1
2	2	1
-1	-1	-1
-2	-1	-1
3	0	1

- Epoch 1
- Bias input: $x_0=1$, weight = w_0
- Initial weights: $w_0=0$, $w_1=0$, $w_2=0$
- Learning rate: $\eta=1$
- $\text{net}=w_0x_0+w_1x_1+w_2x_2$
- Update rule:
- $w_i^{\text{new}} = w_i^{\text{old}} + \eta \cdot y \cdot x_i$

x1	x2	y (Actual)	$\text{net}=w_0x_0+w_1x_1+w_2x_2$	f(net) {1 if $\text{net} \geq 0$, -1 if $\text{net} < 0$ }	Change in Weights ($\Delta w_0, \Delta w_1, \Delta w_2$)	Updated Weights (w_0, w_1, w_2)
1	1	1	0	1	(0, 0, 0)	(0, 0, 0)
2	2	1	0	1	(0, 0, 0)	(0, 0, 0)
-1	-1	-1	0	1	(-1, +1, +1)	(-1, 1, 1)
-2	-1	-1	-4	-1	(0, 0, 0)	(-1, 1, 1)
3	0	1	2	1	(0, 0, 0)	(-1, 1, 1)

- Epoch 2
- Bias input: $x_0=1$, weight = w_0
- Initial weights: $w_0=0$, $w_1=0$, $w_2=0$
- Learning rate: $\eta=1$
- $\text{net}=w_0x_0+w_1x_1+w_2x_2$
- Update rule:
- $w_i^{\text{new}} = w_i^{\text{old}} + \eta \cdot y \cdot x_i$

x1	x2	y (Actual)	$\text{net}=w_0x_0+w_1x_1+w_2x_2$	$f(\text{net})$ $\eta \cdot y$ {1 if $\text{net} \geq 0$, -1 if $\text{net} < 0$ }	Change in Weights ($\Delta w_0, \Delta w_1, \Delta w_2$)	Updated Weights (w_0, w_1, w_2)
1	1	1	1	1	(0, 0, 0)	(-1, 1, 1)
2	2	1	3	1	(0, 0, 0)	(0, 0, 0)
-1	-1	-1	-3	-1	(0, 0, 0)	(-1, 1, 1)
-2	-1	-1	-4	-1	(0, 0, 0)	(-1, 1, 1)
3	0	1	2	1	(0, 0, 0)	(-1, 1, 1)

- Epoch 2 has ZERO updates
- All samples are correctly classified again
- The model is fully converged
- $w_0=-1$, $w_1=1$, $w_2=1$

